

Module: Corporate Finance & Equity Valuation

School of Economics

University of Cape Town

Capital Budgeting

Objectives

- ❑ Understand what Capital budgeting is about
- ❑ Understand capital investment decisions
- ❑ Understand the classification of investment projects
- ❑ Apply the methods used to evaluate capital projects
- ❑ Set out the advantages and disadvantages of each method
- ❑ Determine the relevant cash flows to be included in the analysis
- ❑ Understand the calculation of taxation and tax allowances in the project cash flows
- ❑ Use Excel spreadsheets to solve applied Capital Budgeting problems

What is Capital Budgeting

- ❑ Capital budgeting is the analysis and evaluation of investment projects that normally produce future economic benefits (NET future cash flows = Cash inflows > cash outflows) over a number of years to increase the value of the firm.
- ❑ In assessing projects that will continue for a number of years, we have to consider the entire project's life before making a decision as to whether or not we should go ahead with the project. The financial cash **outlay will normally be upfront, while the benefits** (future cash flows) **will usually be reaped over a number of years.**
- ❑ Often the decision involves choosing between project options and therefore the projects have to be ranked in order of preference.

Capital Investment decisions

- Involves the decision to invest in a project (initial cash outflow) that will result in long term future cash inflows and outflows.
- The investment decision is always taken FIRST, before the financing decision.

□ Capital Budgeting Process

1. Strategic plan (Purchase assets, launch new product, build a new plant, enter a new market and buying a business)
2. Proposal generation
3. Review and analysis
4. Decision making
5. Implementation
 - After the project has been approved the cash expenditures are made
6. Follow up (post project audit)
 - The annual results are monitored and the actual annual actual cash inflows and outflows are compared to the estimated. This will result in appropriate action to be taken.

Classification (types) of Investment Projects

- ❑ Whether a firm can undertake a project will to a large extent be determined by the importance of the project to the existing operations, availability of capital, types of finance to be used.
- ❑ Note that: The availability of funds for capital expenditures will affect the firm's capital budgeting decision. If the firm has unlimited funds then all projects that will increase the value of the firm (have positive net present value) will be accepted and implemented. ^{NPV}
- ❑ The opposite applies if the firm experiences capital rationing. The firm will isolate and accept projects that are high on the priority list and in accordance with the budget set by management.

Examples of projects:

- ❑ Replacement of assets – implies that the firm intends to maintain its existing level of operations (production). The new machine may result in improved production operations, decreasing costs and increasing profits.
- ❑ Expansion : implies that the firm intends to increase its productive capacity to increase the sales of existing products or to diversify into new product lines. Globalization has increased competition, hence, forcing firms to diversify into new product lines in order to remain competitive.
- ❑ Other (IT, Pollution control, Corporate social investment)

❑ Mutually Exclusive vs. Independent Projects

- ✓ i.e. **Alternatives (one or the other) Vs two separate investment decisions.**

❑ Divisible and non-divisible projects

- ✓ **Divisible projects can be split into various parts that may or not have to be done in sequential order**
- ✓ **Indivisible project cannot be broken up into smaller parts and must therefore be undertaken in its entirety.**

not
asked



Capital Budgeting Techniques

☐ Net Present Value ^{NPV} NB! 15-20 marks

☐ Internal Rate of Return ^{IRR}

☐ Modified Internal rate of return (MIRR)

☐ Ordinary or none discounted Payback

☐ Discounted Payback

☐ Accounting Rate of Return ^{ARR}

☐ Profitability Index (PV Index or Benefit-Cost Ratio)

Some principles of capital budgeting

- ❑ Use incremental NET FUTURE cash flows (inflows and outflows) ONLY from project (only what changes in the future, not current Cash flows)
- ❑ Cash flows after tax; Cost of Capital after tax
- ❑ Ignore sunk costs (not relevant)- they are not relevant and cannot influence future decisions
- ❑ Ignore allocated (shared manufacturing costs) costs
- ❑ Ignore finance charges such as interest expense (already in WACC)
- ❑ Include all opportunity costs
- ❑ Include working capital investments and incremental working capital investments
- ❑ Evaluate the effect on profit ^{of} ~~on~~ current products
- ❑ Consider alternatives
- ❑ Finance versus investment decision *always taken first*

BEGINNING of project

- ❑ Purchase price (Purchase price less trade discount add any additional costs such as transport costs, installation costs and trial run costs)
- ❑ Sales price of old (existing) machine
- ❑ Tax on sale (year 1)
- ❑ Working capital requirements and changes

ANNUAL Operating cash flows (during the life of the project)

- ☐ Incremental cash flows
- ☐ Add back depreciation and Interest expense to NET OPERATING PROFIT (accrual) = NET OPERATING CASH FLOWS
- ☐ NET operating cash flows after tax
- ☐ Ignore sunk costs
- ☐ Include all opportunity costs
- ☐ Include effects on existing product lines (competitive or substitutes)
- ☐ Ignore allocated (shared) costs
- ☐ Ignore Interest expense (financing costs) *- already included in WACC*
- ☐ Incremental (changes) working capital

not covering

END of project

❑ Selling price (residual value)

❑ Return of working capital to normal as cash inflow at end of project

❑ Taxation effects of Recoupments / Scrapping allowance *appear in tax schedule*

- *compare residual value & tax value*
- *For you to see recoupments: resid. > tax val.*
- *Residual value goes through NPV schedule*

Normal tax

- ❑ Check time lag (Tax is always due at the end of the year)
- ❑ Section 12 C Wear and tear allowance (normal tax rules)
- ❑ Scrapping allowance (tax value > residual value: results in lower taxable income, lower tax expense, lower cash outflows)
- ❑ Recoupments (tax value < residual value: results in higher taxable income, higher tax expense, higher cash outflows)

Qualitative factors

- ❑ Objective: Maximising shareholder wealth
- ❑ Consider clients, competitors, resources
- ❑ Other objectives: Entering specific markets, environmental requirements (more relevant today), strategic issues e.g. employee safety, health, morale, product quality.

Costs related to Capital Budgeting

❑ Opportunity costs

- Opportunity cost is the potential benefit that is lost or sacrificed when the selection of one course of action makes it necessary to give up a competing course of action.

❑ Sunk costs

- A sunk cost is a cost that was incurred in the past and **does not influence any decision made now or in the future.**

Capital appraisal techniques

Example 1

Example 1 will be used to illustrate the:

-  ^{some of PV = benefits} ○ Net Present Value (NPV), - only accept projects with positive NPV
- Non Discounted Payback Period,
- Discounted Payback Period,
- Accounting Rate of Return (ARR),
- Internal Rate of Return (IRR),

Example 1 continued

Project information:

Initial Investment	R12 million			
Useful life	4 years			
WACC	20%			
Residual Value	Nil			
Year	Accounting Profit	+ Depreciation	= None Discounted Cash flows	Discounted Cash flows
1	R2 million	R3 million	R5 million	R4.17 million
2	R1 million	R3 million	R4 million	R2.78 million
3	R3 million	R3 million	R6 million	R3.47 million
4	R2 million	R3 million	R5 million	R2.47 million
Total	R8 million		R20 million	R12.83 million

NPV Method

Year	0	1	2	3	4
Initial investment	-12				
Cash flows		5	4	6	5
Discount factors@20% ^{WACC}		$\frac{5}{(1+20\%)^1}$ 0.8333	$\frac{4}{(1+20\%)^2}$ 0.6944	$\frac{6}{(1+20\%)^3}$ 0.5787	$\frac{5}{(1+20\%)^4}$ 0.4823
PV of cash flows ^{CF} (net CF)		4.17	2.78	3.47	2.41
Sum of PVs of cash flows	12.83 ⁻¹²				
NPV	0.83				

← positive, ∴ will create value for shareholders

Recommendation: The project should be accepted because it has a positive NPV

Disc factors : $(1+i)^n$
 PV CF : $\frac{FV}{(1+i)^n}$

Application of the NPV method

□ NPV = sum of Future Cash flows discounted at the cost of capital ^{WACC} less the ^{Initial investment} Cost of the Project

o If NPV is positive, accept the project - adds / creates value

o If NPV is negative, reject the project - destroys value

□ A positive NPV means that the project is creating value, i.e. the return that the project generates, exceeds the cost of financing those returns.

□ The advantage of the NPV method is that it takes into account the time ^{TVM} value of money

Non Discounted Payback Period (no TVM)

- Advantages

- Easy to use
- Useful in sifting process – ranking projects
- Widely used in practice
- Indicates effect on liquidity
- Risk indicator – how long the investment cash flow is at risk

Disadvantages

- It does not take into account the time value of money ^{TVM}
- It favors high cash flows early in project life
- It ignores cash flows after payback period
- It is biased against long term projects (liquidity risk increases)

Under what circumstances is this method appropriate

- Liquidity is poor
- Uncertainty about future
- Cash flows of projects uncertain

Application of the Non Discounted Payback Period

Initial Investment	-12
Less. Year 1 Net cash flows	<u>5</u>
Balance	-7 > 4
Less. Year 2 Net cash flows	<u>4</u>
Balance	-3 < 6 ∴ stop here

payback period :
 $2 + \frac{3}{6} = 2.5$ yrs
(how long it'll take
to recover initial
investment)

We stop doing the calculation the moment the balance is less than the next Net Cash flow. In the case above, the balance of –R3 is lower than the Net cash flow in year 3, (i.e. $-3 < 6$, thus we have to stop the calculation and express the balance as a proportion of the next Net cash flow, since the firm would be able to recover the remaining balance within the next year

Thus, the payback period would be $2 + (3/6) = 2.5$ years. So this means that it will take 2.5 years to recover the initial investment. If I wanted the answer in years and months $\rightarrow 2 + [(3/6)*12] = 2$ years 6 months. I can as well do a calculation for years, months and days.

Recommendation: If benchmark is 2 years, the project will be rejected since $2.5 > 2$

Meaning, that it will take longer to recover the cost of the investment than is recommended by the firm.

Discounted Payback Period

- ❑ This method attempts to overcome one of the problems of the none discounted or ordinary payback method by discounting the cash flows.
- ❑ Discounted Payback Period is the length of time required for the Discounted Net Future cash inflows to pay back the initial investment using an appropriate discount rate.
- ❑ Advantages
 - Easy to use
 - Useful in sifting process
 - Takes into account the time value of money TVM
- ❑ Disadvantages
 - It favors high cash flows early in project life
 - It ignores cash flows after payback period
- ❑ Under what circumstances is this method appropriate
 - Liquidity is poor
 - Uncertainty about future
 - Cash flows of projects uncertain

Application of the ~~Non~~ Discounted Payback Period

Initial Investment	-12
Less. Year 1 Net cash flows	<u>4.17</u>
Balance	-7.83 > 2,78
Less. Year 2 Net cash flows	<u>2.78</u>
Balance	-5.05 > 3,47
Less. Year 3 Net Cash flow	<u>3.47</u>
Balance	-1.58 < 2,41 ∴ stop

payback period :

$$3 + \frac{1,58}{2,41} = 3,66 \text{ yrs}$$
$$= 3 \text{ yrs} + 7,87 \text{ months}$$
$$= 3 \text{ yrs} + 7 \text{ mo} + 26 \text{ days}$$

We stop doing the calculation the moment the balance is less than the next Net Cash flow. In the case above, the balance of –R1.58 is lower than the Net cash flow in year 4, (i.e. - 1.58 < 2.41. Thus, the payback period would be $3 + (1.58/2.41) = 3.66$ years. So this means that it will take 3.66 years to recover the initial investment. If I wanted the answer in years and months -> $3 + [(1.58/2.41)*12] = 3$ years and 7,86722 months which is approximately 3 years and 8 months. Which is also approximately 3 years 7 months + $(0.86722*30) = 3$ years, 7 months and 26 days if I wanted the calculation in years, months and days.

Recommendation: If benchmark is 4 years, the project will be accepted since $3.66 < 4$

Accounting Rate of Return (ARR)

- ❑ $ARR = \text{Average profit} / \text{Average Investment}$
- ❑ Where $\text{Average Investment} = (\text{At cost} + \text{Residual Value}) / 2$
- ❑ ARR measures the effectiveness with which management is using the company's assets.

Disadvantages

- Ignores time value of money. ^{TVM}
- Projects with high income in early years and low income in later years will give same result as projects with low income in early years and high income in later years.
- Uses Accrual Accounting income rather than cash flows.
- May lead to sub-optimal decision-making by managers who are evaluated based on ROI [Net profit after tax / Total assets].

ARR

From the example:

$$\text{ARR} = (\text{average profit} / \text{average investment}) * 100$$

$$\text{Average profit} = \overset{\text{Accounting profit}/4}{(2 + 1 + 3 + 2)/4} = 2$$

$$\text{Average investment} = \overset{(\text{At cost} + \text{resid})/2}{(12 + 0)/2} = 6$$

$$\text{Therefore, ARR} = \overset{\text{Ave profit}/\text{Ave investment}}{(2/6)} * 100 = 33\%$$

If the benchmark is 30%

Recommendation: Accept the project since $\text{ARR} > \text{Benchmark}$

.i.e. $33\% > 30\%$

Internal Rate of Return (IRR)

❑ IRR is the Discount rate that will result in a zero NPV (.i.e. it is that discount rate that if used to discount the future cash flows, will result in Cash inflows being equal to cash outflows). (Actual rate earned)

❑ Advantages

- Cash flows; Time value of money; Percentage (comparison)
- Don't need to know WACC beforehand.

Disadvantages

- Re-investment assumption @ IRR (NPV @ WACC)
- Results of IRR & NPV may conflict. Prefer re-investment at cost of capital because:
- It gives a more conservative estimate
- Cost of capital remains constant, compared to IRR where re-investment rates differ for different projects
- With unlimited finance, projects with $IRR > WACC$ are accepted, therefore NPV is best.
- IRR is a percentage (no monetary value)
- Not appropriate for mutually exclusive projects of different sizes (IRR & NPV give different rankings)
- Not appropriate for projects with unconventional cash flows or where timing of cash flows differ (this results in more than one IRR)

* won't be asked to do

Internal Rate of Return (IRR) Trial and Error method

Year	Net Cashflows	20%	21%	22%	23%	24%
	R'million'	$FV/(1+i)^n$	$FV/(1+i)^n$	$FV/(1+i)^n$	$FV/(1+i)^n$	$FV/(1+i)^n$
1	5	4.16667	4.13223	4.09836	4.06504	4.03226
2	4	2.7778	2.73205	2.68745	2.64393	2.60146
3	6	3.47222	3.38684	3.30424	3.22430	3.12419
4	5	2.41127	2.33254	2.25700	2.18449	2.11487
Sum of PVs		12.82793	12.58367	12.34705	12.11776	11.87278
Less. Cost		-12	-12	-12	-12	-12
NPV		0.82793	0.58367	0.34705	0.11776	-0.12722
					last +ve NPV	

Interpolation: To calculate the IRR you take the **discount rate** associated with the **last positive NPV** and add the proportion associated with a division of the last positive NPV by the sum of the NPVs associated with the last positive NPV and the absolute value of the first negative NPV, see below.

$$23\% + \left[\frac{\text{NPV}}{\text{NPV} + \text{next NPV (abs. no.)}} \right] = 23.48\% = \text{IRR}$$

Advanced Examples in capital budgeting

Example 2 Solution (workings)

^{w1} Working 1 (Calculation of accounting depreciation)

Assume a straight line method.

Depreciation = (At cost minus residual value)/useful life
of the asset.

$$\text{Depreciation} = (500\,000 - 50\,000)/4 = 112\,500$$

Advanced Examples in capital budgeting

Example 2 Solution (workings)

^{w2}
Working 2 (Calculation of Recoupment/scrapping allowance)

At Cost R500 000.

Depreciation based on sec 12 C is 20% per annum

Therefore, sec. 12 C depreciation = $0.2 \times 500\,000 = 100\,000$ per year

Thus, accumulated depreciation based on sec. 12C over 4 years is R400 000

Determination of recoupment or scrap

At cost	500 000
Less. Accumulated depreciation based on sec. 12C	<u>400 000</u>
Tax Value	100 000

vs resid value R50 000
Tax value > resid
∴ scrapping allowance

Recoupment = Residual value minus Tax value

If Residual > Tax value = Recoupment (this increases taxable income, hence should be recorded as a plus, that is a positive in the taxation schedule)

If Residual value < Tax value = scrap allowance (this decreases or reduces taxable income, hence should be recorded in the taxation schedule as a minus, that is a negative)

In this case, Residual value = R50 000 and the Tax value = R100 000

Therefore, since Residual value < Tax value, thus re record a scrapping allowance

Scrape allowance = Residual value – Tax value = $50\,000 - 100\,000 = -50\,000$ ^{w2}, so the moment the difference between the residual value and Tax value is negative, know that you have scraping allowance, and treat that accordingly in the taxation schedule (see taxation schedule)

Advanced Examples in capital budgeting

Example 2 Solution (workings)

^{W3} Working 3 (Taxation schedule- calculation of the taxable income).

	1	2	3	4
Net operating profit After DEDUCTING Depreciation & interest but before tax	200 000	210 000	230 000	240 000
Add. Interest expense	30 000	30 000	30 000	20 000
Add. Depreciation (accounting depreciation) W1	<u>112 500</u>	<u>112 500</u>	<u>112 500</u>	<u>112 500</u>
Net Operating cash flows before Tax and other adjustments (EBITDA)	342 500	352 500	372 500	372 500
Other Adjustments				
Less. Sec. 12C Allowance W2	(100 000)	(100 000)	(100 000)	(100 000)
Less. Scrapping allowance W2				(50 000)
Taxable Income	242 500 <i>x 28%</i>	252 500	272 500	222 500
Taxation (28% of Taxable Income)	67 900	70 700	76 300	62 300

Advanced Examples in capital budgeting

Example 2 Solution (workings)

^{w4} Working 4 (Conversion of Accrual Operating Profits to Gross OPERATING CASH INFLOWS before Tax).

Year	1	2	3	4
Net operating profit After DEDUCTING Depreciation & interest but before tax	200 000	210 000	230 000	240 000
Add. Interest expense	30 000	30 000	30 000	20 000
Add. Depreciation W1	<u>112 500</u>	<u>112 500</u>	<u>112 500</u>	<u>112 500</u>
Gross operating Cash Flows (EBITDA)	342 500	352 500	372 500	372 500
Residual value				50 000
Working capital recovered				60 000
Taxation (see taxation schedule) W3	<u>(67 900)</u>	<u>(70 700)</u>	<u>(76 300)</u>	<u>(62 300)</u>
Net operating cash flows after Tax	274 600	281 800	296 200	420 200

Same as W3

Advanced Examples in capital budgeting

Example 2 Solution Part 1

Calculation of the ordinary Payback period (for the machine)

	Cash flows
Cost of the Machine	(500 000)
Less. Year 1 Cash flow W4	274 600
Balance	(225 400)
Now the balance is less than the next cash flow of R281 800, so we need to proceed as below	

Thus, the payback period = $1 + \frac{225\,400}{281\,800} = 1.8$ years
or 1 year, 9 months and 18 days.

To convert to years and months = $1 + (225\,400/281\,800) * 12$

To convert to years, months and days = $1 + (225\,400/281\,800) * 12$

Which is = 1 year + 9,5982966648 months

Which is = 1 year, 9 months + (0,5982966648*30)

Which is = 1 year, 9 months and 18 days.

Advanced Examples in capital budgeting

Example 2 Solution Part 2

The NPV schedule. (since NPV is positive, accept the project.)

	0	1	2	3	4
Initial Cost - Machine	(500 000)	} spending			
Working Capital	(60 000)				
Gross operating Cash Flows <i>W4 (EBITDA)</i>		342 500	352 500	362 500	372 500
Residual					50 000
Taxation <i>W3</i>		<u>(67 900)</u>	<u>(70 700)</u>	<u>(73 500)</u>	<u>(62 300)</u>
<i>NOCF</i> Net Operating cash flows <i>W4</i>	(560 000)	274 600	281 800	296 200	420 200
Discount factors at a WACC of 15%	<u>1</u>	<u>0.8696</u>	<u>0.7561</u>	<u>0.6575</u>	<u>0.5718</u>
Pvs of Net operating Cash flows <i>(Net CF)</i>	(560 000)	238 783	+ 213 081	+ 194 756	+ 240 251
Sum of PVs of Net operating Cash flows	<u>886 871</u>				
NPV (.i.e. 882 137 minus 560 000)	326 871				

Scrapping allowance & Recoupment

Issues to remember:

Tax Value = At cost less Accumulated depreciation based on section 12C.

Scrapping allowance

Scrapping allowance is when the Tax Value > Residual Value.

The effect scrapping allowance is that it decreases Taxable income ,
decreases Tax expense, decreases cash flows

Recoupment

Recoupment is when the Tax Value < Residual Value.

The effect of recoupment is that it increases Taxable income, increases Tax
expense, increases cash flow

Advanced Examples in capital budgeting

- Example 3 Solution (workings)

^{w1}Working 1 (Initial cost of the Machine)

Price before discount	3 200 000
Discount	(200 000)
Installation cost	400 000
Total cost of the machine	3 400 000

^{w2}Working 2 (Residual value new machine - 10% of total cost)

Residual value = $10\% \times 3\,400\,000$

Residual value (new machine) = **340 000**

Advanced Examples in capital budgeting

- Example 3 Solution (workings)

^{w3}
Working 3 (Calculation of section 12C accumulated depreciation on new Machine)

Depreciation in year 1 (sec. 12C) 40%* ^{full @ cost value} 3 400 000	1 360 000
Depreciation in year 2 (sec. 12C) 20%* 3 400 000	680 000
Depreciation in year 3 (sec. 12C) 20%* 3 400 000	<u>680 000</u>
Accumulated depreciation over 3 years	2 720 000

Advanced Examples in capital budgeting

- Example 3 Solution (workings)

Working 4 (scrapping or recoupment for the new Machine)

At cost	3 400 000
Less. Accumulated depreciation (sec. 12C)	<u>(2 720 000)</u>
Tax value	680 000 > 340 000

Residual value for new Machine (see working 2) = 340 000
 goes to NPV schedule *∴ scrapping allowance*

Since Tax Value > Residual value, we value a scrapping allowance

Scrapping allowance = Residual minus Tax value

Scrapping allowance = 340 000 – 680 000

Scrapping allowance (new machine) = -340 000 *goes to tax schedule*

Advanced Examples in capital budgeting

- Example 3 Solution (workings)

^{WS} Working 5 (scrapping or recoupment for the old Machine)

Tax Value (see note 3)	0
Residual value (note 3)	<u>100 000</u>
	<i>∴ Recoupment = R100k</i>

Since Residual value > Tax value, we have Recoupment

Recoupment value = Residual minus Tax value

Recoupment value (old machine) = 100 000 – 0

Recoupment value (old machine) = **100 000**

Advanced Examples in capital budgeting

- Example 3 Solution (workings)

^{w6}Working 6 (Calculation of ^{NOCF}NET Operating cash flows)

Year	1	2	3
Net operating profits (NOP)	1 800 000	2 000 000	2 400 000
Add. Depreciation (accounting depreciation)	1 020 000	1 020 000	1 020 000
Add. interest	<u>60 000</u>	<u>60 000</u>	<u>60 000</u>
^{NOCF} Net operating cash flow	2 880 000	3 080 000	3 480 000

Advanced Examples in capital budgeting

- Example 3 Solution (workings)

^{W7}Working 7 (Taxation Schedule)

Year	1	2	3
Net operating cash flows ^{NOCF} W6	2 880 000	3 080 000	3 480 000
Less. Depreciation sec. 12C W3	<u>(1 360 000)</u>	<u>(680 000)</u>	<u>(680 000)</u>
Less. Lost profits (notes)	(80 000)	(80 000)	(90 000)
Less. Fixed costs (notes)	(650 000) ^{+5%}	(682 500) ^{+5%}	(716 625)
Less. Opportunity cost (notes) ^(rent) 10k x 12	(120 000)	(120 000)	(120 000)
Less. Scrapping allowance (new) W4			(340 000)
Add. Recoupment (old) W5	100 000		
Taxable income	770 000	1 517 500	1 533 375
Taxation @28%	215 600	424 900	429 345

Advanced Examples in capital budgeting (Example 3 Solution (NPV calculation))

Year	0	1	2	3
Initial cost W1	(3 400 000)			
<i>ignore</i> Sunk costs (background information)	-			
<i>NOCF</i> Net operating cash flows W6		2 880 000	3 080 000	3 480 000
Residual value (old) W5	100 000			
Residual value (new) W4				340 000
Working capital - Inventory <i>Cumulative numbers</i>	<i>starting with</i> (500 000) <i>spending initially</i>	<i>Cumulative no.</i> (50 000)	<i>recovered</i> 100 000	<i>500 + 50 - 100</i> 450 000 <i>sum = 0</i>
- Debtors		(350 000)	<i>recovered</i> 30 000 <i>debtors paid</i>	<i>still to recover</i> 320 000 <i>sum = 0</i>
Opportunity cost- project		(80 000)	(80 000)	(90 000)
Fixed Costs- project		(650 000)	(682 500)	(716 625)
Opportunity cost rent (12 x 10 000)		(120 000)	(120 000)	(120 000)
Taxation W7		(215 600)	(424 900)	(429 345)
<i>NOCF</i> Net Operating Cash flows	(3 800 000)	1 414 400	1 902 600	3 234 030

present *future* →

Advanced Examples in capital budgeting (Example 3 Solution (NPV calculation))

Example 3 Continued NPV calculation

Year	0	1	2	3
Net Operating Cash flows	(3 800 000)	1 414 400	1 902 600	3 234 030
Discount factors @15%		<u>0,8696</u>	<u>0,7561</u>	<u>0,6575</u>
PVs of cash flows		1 229 913	1 438 639	2 126 427
Sum of PVs of Cash flows	<u>4 794 979</u>			
NPV (4 794 979 minus 3 800 000)	994 979			

Decision: Accept the project since it yields a positive NPV

Advanced Examples in capital budgeting

Example 4 Solution (workings)

^{w1}Working 1 (calculation of revenues)

$$\text{Revenue} = 300 \times 10\,000 = 3\,000\,000$$

^{w2}Working 2 (Calculation of cost of sales)^{cos}

	workings	
Material fabric	3m x R28.5	85.50
Wood	3.5m x R32.00	112.00
Labour	1.75 x R22	38.50
Total/unit		236.00
Total Cost of sales	236.00 x 10 000	2 360 000

Advanced Examples in capital budgeting

Example 4 Solution (workings)

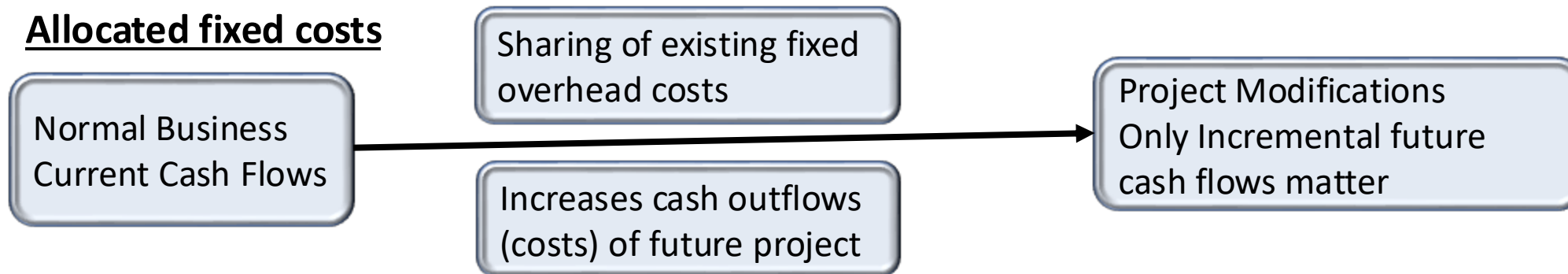
^{w3}Working 3 (calculation of accounting depreciation)

Depreciation = (At cost – residual value)/useful life

Depreciation = $(800\,000 - 200\,000)/5 = 120\,000$

^{w4}Working 4 (Calculation of fixed costs). Note the following.

Allocated fixed costs



Total fixed costs (given)	200 000
Less. Depreciation	(120 000)
Less. Allocated fixed costs	<u>(40 000)</u>
Expected relevant fixed costs of the project	40 000

Advanced Examples in capital budgeting

- Working 5 (calculation of Incremental working capital. (working capital is 10% of sales revenue)

	0	1	2	3	4	5
Sales Revenue		3 000 000	3 127 500	3 260 419	3 398 987	3 543 443
Working Capital (10%)		300 000	312 750	326 042	339 898.7	354 344.3
Incremental working capital	-300 000	+ -12 750	+ -13 292	+ -13 857	+ -14 446	= 354 345

Things to note:

- The working capital numbers are cumulative, thus you need to calculate the changes as the changes represent additional expenditure on working capital
- The first amount in working capital is spend at the beginning of the project, thus why we have -300 000 under year 0.
- So after noting the above, all the changes are then lagged in such as way that they end in the year prior to the end of the project as you can see above.
- By the end of the project the company should be able to recover all the money that is expected to be invested in working capital over the project life, thus the 354 345 was got by adding the incremental working capital numbers as below

Working capital recovered = $(300\,000 + 12\,750 + 13\,292 + 13\,857 + 14\,446) = 354\,345$

Advanced Examples in capital budgeting

^{w6}
Working 6 (calculation of sec. 12C depreciation, 40%, 20%, 20%, 20%)

Year 1	40% x 800 000 = 320 000
Year 2	20% x 800 000 = 160 000
Year 3	20% x 800 000 = 160 000
Year 4	20% x 800 000 = 160 000

^{w7}
Working 7 (calculation for recoupment/scrapping allowance)

At cost	800 000
Accumulated depreciation sec. 12 C	<u>800 000</u>
Tax Value	0

Residual value = 200 000

Recoupment value = Residual value minus Tax Value

Recoupment value = 200 000 – 0

Recoupment value = **200 000**

Advanced Examples in capital budgeting

Working 6 Taxation schedule (Calculation of taxable income)

Year	1	2	3	4	5
Revenue/Sales W1 (illustration)	3 000 000	3 127 500	3 260 419	3 398 987	3 543 443
Cash cost of sales W2 (illustration)	(2 360 000)	(2 460 300)	(2 564 863)	(2 673 869)	(2 787 509)
Sec12 C allowance W6	(320 000)	(160 000)	(160 000)	(160 000)	-
Fixed costs for project W4	(40 000)	(40 000)	(40 000)	(40 000)	(40 000)
Opportunity costs – Rent (notes)	(36 000)	(36 000)	(36 000)	(36 000)	(36 000)
Recoupment W7					200 000
Opportunity cost- of existing residual value (notes)	(120 000)				
Taxable income	124 000	431 200	459 556	489 117	879 935
Taxation @ 29% (notes)	35 960	125 048	133 271	141 844	255 181

Advanced Examples in capital budgeting (NPV schedule)

	0	1	2	3	4	5
Initial cost (modification)	(800 000)					
Opportunity cost – existing res value	(120 000)					
Sunk cost- Marketing research		ignore always show, but = 0 (won't change our decision)				
Sales revenue W1		3 000 000	3 127 500	3 260 419	3 398 987	3 543 443
Cost of sales W2		(2 360 000)	(2 460 300)	(2 564 863)	(2 673 869)	(2 787 509)
Taxation W6		(35 960)	(125 048)	(133 271)	(141 844)	(255 181)
Fixed costs W4		(40 000)	(40 000)	(40 000)	(40 000)	(40 000)
Working capital W5	(300 000)	(12 750)	(13 292)	(13 857)	(14 446)	354 345
Opportunity cost rental		(36 000)	(36 000)	(36 000)	(36 000)	(36 000)
Residual Value (notes)						200 000
Net Annual future Cash flow	(1 220 000)	515 290	452 860	472 428	492 827	979 098
Discount factors @16%	<u>1</u>	<u>0.8621</u>	<u>0.7432</u>	<u>0.6407</u>	<u>0.5523</u>	<u>0.4761</u>
Pvs of Cash Flows	(1 220 000)	444216	336549	302665	272184	466161
Sum of PVs of cash flows	<u>1821774</u>					
NPV (1 821 774 – 1 220 000)	601774					

Advanced Examples in capital budgeting (calculation of Payback period)

Initial cost of modification	
Initial cost	(800 000)
Less. Year 1 Cash flow	515 290
Balance	(284 710) < 452 860
	∴ stop here
Payback = 1 + (284 710/452 860) = 1,63 years	